

This document tests equation extraction.

An inline equation: Einstein's mass-energy equivalence relation, $E = mc^2$, establishes that energy equals mass times the speed of light squared, revealing the fundamental convertibility between mass and energy.

~~A display equation~~ The Pythagorean theorem states that in a right triangle, the sum of the squares of the two legs equals the square of the hypotenuse:

$$a^2 + b^2 = c^2 \tag{1}$$

~~Another display equation without a label:-~~

~~$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$~~

~~The value~~ where a and b represent the lengths of the two legs, and c represents the length of the hypotenuse. This relationship, referenced by ~~1~~is fundamental, is fundamental to Euclidean geometry and underlies numerous results in mathematics and physics.

The Gaussian integral evaluates to a closed form:

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

demonstrating that the integral of the Gaussian function from zero to infinity equals $\sqrt{\pi}/2$.